

Multiple Linear Regression

This model, also known as **linear regression with multiple features**, is an extension of the model from Week 1. It allows us to use more than one input feature to make more accurate predictions.

Motivation 🤔

Instead of predicting a house's price based only on its size, we can use additional information to improve our model.

- **Single Feature (Week 1):** x = size
 - **Multiple Features (Week 2):**
 - x_1 = size
 - x_2 = number of bedrooms
 - x_3 = number of floors
 - x_4 = age of home
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New Notation 📝

With multiple features, we need to update our notation to handle them.

- x_j : Denotes the **j-th feature** in our feature list (e.g., x_2 is the number of bedrooms).
 - **n**: The **total number of features** (in our example, $n=4$).
 - $\vec{x}^{(i)}$: Represents the **i-th training example**. This is now a **vector**, or a list of numbers, containing all the feature values for that example.
 - For example, $\vec{x}^{(2)} = [1416, 3, 2, 40]$.
 - $x_j^{(i)}$: The value of the **j-th feature** in the **i-th training example**.
 - For example, $x_3^{(2)} = 2$ (the number of floors in the second house).
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The Updated Model Equation

The model is extended from a simple line to an equation that incorporates all n features.

- **Model with n features:**
$$f_{\vec{w},b}(\vec{x}) = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

To write this more compactly, we use vector notation and the **dot product**.

1. Define Parameter and Feature Vectors:

- The parameters w are now a vector: $\vec{w} = [w_1, w_2, \dots, w_n]$
- The features x for a single example are a vector: $\vec{x} = [x_1, x_2, \dots, x_n]$
- b remains a single number (a scalar).

2. **Use the Dot Product:** The dot product of $\vec{w}$ and $\vec{x}$ is defined as:

$$\vec{w} \cdot \vec{x} = w_1x_1 + w_2x_2 + \dots + w_nx_n$$

3. **The Final, Compact Model:**

$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$